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PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Autonomous Drone Fleet Routing Management System

Team ID: DSCPP-III-2026-T154 **Academic Session:** 2026–27 **Semester:** 3rd (CSE)	**Domain:** Autonomous Systems / Fleet Logistics
Team member 1 (Team Lead) *Jain, Saksham : 2510380116 : 2510380116@geu.ac.in* **Focus:** System Architecture, 3D Graph Corridors & A^* Pathfinding Routing Engine	
Team member 2 *Jiya : 2510370376 : 2510370376@geu.ac.in* **Focus:** Min-Heap Priority Queue Scheduling, Fleet Telemetry Hash Map & Validation	
Team member 3 *Bhinder, Tamanna : 2510380219 : 2510380219@geu.ac.in* **Focus:** Aerodynamic Wind Physics, Energy Expenditure Modeling & Testing Support	

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

The primary objective of the **Autonomous Drone Fleet Routing Management System** is to architect and implement an algorithmic dispatching and trajectory optimization engine in modern C++17[cite: 1, 4]. Last-mile logistics remains congested and carbon-intensive; while autonomous Unmanned Aerial Vehicles (UAVs) offer rapid point-to-point delivery, multi-drone operations face bounded lithium-polymer battery capacities, non-linear aerodynamic power expenditure, customer deadlines, and ambient wind disturbances[cite: 1, 4].

In Phase-II, approximately **70% of the end-to-end system has been realized**[cite: 1]. The engine integrates: (1) an Earliest-Deadline-First (EDF) Min-Heap Priority Queue for $O(\log N)$ order ingestion[cite: 1, 4]; (2) an Adjacency-List-based 3D Airway Corridor Graph ($O(|V| + |E|)$) with dynamic edge costs factoring in payload weight and atmospheric wind vectors ($\vec{v}_g = \vec{v}_a + \vec{v}_w$)[cite: 1, 4]; (3) an $O(1)$-average Hash Map telemetry hub for UAV tracking[cite: 1, 4]; (4) an A^* Search routing core with an admissible Euclidean energy heuristic[cite: 1, 4]; and (5) a corridor visualization module colorizing active routes in Green and blocked hazards in Red[cite: 1].

Updated Project Approach and Architecture (2 pts)

The system is built as a modular Object-Oriented C++17 application compiled via the GNU Compiler Collection (GCC / g++) toolchain[cite: 1, 4]:

• **Order Ingestion & Scheduling Layer:** Requests are enqueued into a Min-Heap priority queue (`std::priority_queue` with `operator>`), guaranteeing deliveries with earliest deadlines are scheduled first in $O(\log N)$ time[cite: 1, 4].

• **Fleet Telemetry Subsystem:** Managed via `std::unordered_map<std::string, Drone>` to ensure $O(1)$-average telemetry updates for State-of-Charge (SoC), payload capacity, and availability[cite: 1, 4].

• **3D Airway Routing Core:** Flight corridors are represented as a directed graph using an adjacency list[cite: 1, 4]. Dynamic edge weights factor in parasite drag, mass, and wind fields ($P = P_{\text{base}} + k_{\text{mass}} M + k_{\text{drag}} v_a^3$)[cite: 4]. Optimal trajectories are computed using A^* search with admissible lower-bound energy heuristics[cite: 1, 4].

• **Corridor Telemetry Visualizer:** Corridors track discrete states (`OPEN`, `RUNNING_MISSION`, `BLOCKED_RESTRICTED`), displayed in console radar (Green/Red) and exported to an interactive HTML/SVG canvas (`airspace_map.html`)[cite: 1].

Tasks Completed (7 pts)

Task Completed	Team Member
• Developed modular C++17 architecture and 3D vector algebra (`Vector3D`)[cite: 1, 4]. • Built sparse 3D Airway Graph adjacency list[cite: 1, 4]. • Implemented A^* search routing engine with admissible Euclidean energy heuristic and no-fly zone corridor pruning[cite: 1, 4]. • Designed Earliest-Deadline-First (EDF) Min-Heap priority queue for order scheduling ($O(\log N)$ push/pop)[cite: 1, 4]. • Implemented $O(1)$-average Hash Map fleet telemetry registry (`unordered_map`) for UAV SoC and status tracking[cite: 1, 4]. • Formulated wind-vector ground speed ($\vec{v}_g = \vec{v}_a + \vec{v}_w$) and non-linear aerodynamic power expenditure models[cite: 4]. • Implemented visual corridor matrix and ANSI/HTML color-coded radar[cite: 1].	**Saksham Jain** (Lead) **Jiya** **Tamanna Bhinder**

Challenges/Roadblocks (7 pts)

1. **Non-Euclidean Dynamic Edge Weights under Crosswinds:** Unlike static graphs where distance is symmetric, wind causes travel time and energy to vary directionally ($E_{u\to v} \neq E_{v\to u}$)[cite: 4]. *Solution:* Implemented ground velocity projection ($\vec{v}_g = \vec{v}_a + \vec{v}_w$) and evaluated dynamic power draw for each edge relaxation step[cite: 4].

2. **Admissibility of A^* Heuristic:** A naive distance heuristic can overestimate costs under tailwinds, violating A^* admissibility[cite: 4]. *Solution:* Formulated a strict lower-bound heuristic: $h(u) = (d_u/(v_a + \|\vec{v}_w\|_{\max})) \times (P_{\min}/3600)$, guaranteeing globally optimal least-energy paths[cite: 4].

3. **Corridor Pruning & Disconnected Topologies:** Disconnecting corridors due to weather hazards caused unreachable target nodes without alternative paths[cite: 1, 4]. *Solution:* Added graph connectivity sanity checks and alternative bypass corridors[cite: 4].

Tasks Pending (7 pts)

Task Pending (Remaining 30% Scope for Phase-III)	Team Member (Assigned)
• Integration of dynamic mid-air rerouting and conflict resolution when corridor status updates during transit[cite: 1, 4]. • Software-In-The-Loop (SITL) autopilot interface with MAVLink waypoint streaming[cite: 4]. • Comprehensive benchmarking against FIFO baselines for makespan and energy reduction metrics[cite: 1, 4].	**Saksham Jain** (Lead) **Jiya** **Tamanna Bhinder**

Project Outcome/Deliverables (2 pts)

The primary deliverables of the project are:

1. A production-grade, compilable **C++17 autonomous drone fleet routing software suite** built using modular Object-Oriented principles and the GCC toolchain[cite: 1, 4].

2. **Core DSA Dispatch Modules:** An Earliest-Deadline-First (EDF) Min-Heap priority scheduler ($O(\log N)$), an $O(1)$-average Hash Map fleet telemetry registry, and a 3D sparse airway graph adjacency list[cite: 1, 4].

3. **Aerodynamic & Pathfinding Engine:** An A^* search routing engine supporting wind vector calculations, dynamic edge energy weights, and no-fly zone obstacle avoidance[cite: 1, 4].

4. **Interactive Visual Telemetry Interfaces:** An ANSI color-coded console radar map and an automated HTML/SVG export (`airspace_map.html`) displaying running (Green) and blocked (Red) flight corridors[cite: 1].

Progress Overview (2 pts)

Current Completion Status: 70% Completed (Phase-II Target Met)[cite: 1]

• **Completed Core Work (70%):** The end-to-end algorithmic pipeline is fully operational[cite: 1, 4]. Order ingestion, Min-Heap EDF priority queue scheduling, Hash Map telemetry lookups, aerodynamic edge cost computation, A^* pathfinding, battery SoC round-trip feasibility checks, and color-coded corridor topology visualization are completely implemented, verified, and unit-tested in C++17[cite: 1, 4].

• **Ahead of Schedule:** The visualization subsystem (ANSI console radar and automatic interactive HTML/SVG export) was originally planned for Phase-III but was completed early to assist Phase-II demonstration[cite: 1].

• **Remaining Work (30% - Phase-III Scope):** Reserved for the final milestone: (1) MAVLink/SITL simulator telemetry bridges; (2) dynamic mid-air rerouting when corridors close during transit; and (3) formal empirical benchmarking evaluating makespan reduction against baseline algorithms[cite: 1, 4].

Codebase Information (2 pts)

Repository URL: `https://github.com/SakshamJain-2026/Autonomous-Drone-Fleet-Routing`
Primary Branch: `main` | **Build Standard:** Modern C++17 (GCC/g++ 9+)[cite: 1, 4]

Key Commits & Milestones Completed:

- `commit 3b7a12`: Implemented `Vector3D` algebra, wind field vector mechanics, and base types[cite: 4].
- `commit 5c91e4`: Built `AirwayGraph` adjacency list and aerodynamic edge cost functions[cite: 1, 4].
- `commit 7e20d1`: Implemented `OrderScheduler` Min-Heap priority queue and `FleetManager` hash map[cite: 1, 4].
- `commit 9a44b8`: Developed A^* search routing engine with admissible heuristic and obstacle avoidance[cite: 1, 4].
- `commit e105f2`: Added Phase-II interactive CLI, ANSI colored radar, and HTML map generator[cite: 1].

Testing and Validation Status (2 pts)

Test Type	Status	Notes
Min-Heap EDF Order Prioritization	**PASS**	Orders with tighter deadlines correctly sorted to heap root and popped first ($O(\log N)$)[cite: 1, 4].
Dynamic Wind Drag & Cost Computation	**PASS**	Headwind vectors correctly increased transit time and energy consumption dynamically[cite: 4].
A^* Obstacle & Restricted Corridor Pruning	**PASS**	Corridors flagged as `BLOCKED` safely avoided; routes successfully rerouted through bypasses[cite: 1, 4].
Battery SoC Reserve Feasibility Checks	**PASS**	Missions requiring $> 85\%$ round-trip battery or exceeding deadlines are safely rejected[cite: 1, 4].

Deliverables Progress (2 pts)

- **Deliverable 1 (Core C++17 DSA Dispatch Engine):** **COMPLETED (100%)** – Min-Heap priority queue, Adjacency List graph, and Hash Map registry fully functional[cite: 1, 4].
- **Deliverable 2 (Wind Physics & A^* Pathfinding):** **COMPLETED (100%)** – Dynamic aerodynamic drag, energy weights, and restricted corridor bypass validated[cite: 1, 4].
- **Deliverable 3 (Visual Telemetry Radar & HTML Map):** **COMPLETED (100%)** – Real-time console radar and SVG visualizer operational[cite: 1].
- **Deliverable 4 (Dynamic Mid-Air Rerouting & SITL/MAVLink):** **PENDING (Phase-III)** – Scheduled for Phase-III integration[cite: 1, 4].
- **Deliverable 5 (Makespan & Energy Benchmarking Suite):** **IN PROGRESS (Phase-III)** – Baseline dataset collection underway[cite: 1, 4].